THE TRANS $-2/9$ IDENTITY FOR DEGREE-$(2,1)$ POLYNOMIAL CONTINUED FRACTIONS: EMPIRICAL EVIDENCE, AN ALGEBRAIC SIGNATURE, AND THE LIMITS OF STANDARD ASYMPTOTIC METHODS

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Abstract. For an integer polynomial continued fraction (PCF) of partial-numerator degree 2 and partial-denominator degree 1, write $a(n) = a_2n^2 + a_1n + a_0$ and $b(n) = b_1n + b_0$ with $a_2, b_1 \neq 0$, and set the Worpitzky parameter $r := a_2/b_1^2$. A computational survey across approximately 585,000 integer families finds zero counterexamples to the empirical statement that every transcendental-stratum (Trans) limit lying in the negative-Worpitzky interior arises at $r = -2/9$, equivalently at indicial exponents $\{1/3, 2/3\}$ for the standard plus-form recurrence; this is the *Transcendental Ratio Identity* announced in the companion note [1]. The present paper extends that announcement along two axes.

First, we establish the *algebraic signature* of the locus $r = -2/9$: under the alternative minus-form recurrence $y_{n+1} = b(n)y_n - a(n)y_{n-1}$, the characteristic roots are $\mu_\pm = b_1(3 \pm \sqrt{17})/6$, and we prove that, for $a_2 < 0$, $b_1 \neq 0$, $r = -2/9$ is equivalent to the modulus identity $|\mu_+/\mu_-| = (13 + 3\sqrt{17})/4$ (PSLQ-certified at dps = 150 with integer relation $[13, 3, -4]$ on $\{1, \sqrt{17}, \rho\}$). This is the first explicit identification of a quadratic irrational ($\sqrt{17}$) as an algebraic invariant of the Trans property.

Second, we record three negative results that bound the analytic landscape: (i) under the minus-form recurrence the indicial pair $\{1/3, 2/3\}$ corresponds to a compatibility ideal intersecting $r = -2/9$ only at irrational $b_0/b_1 = (27 \pm \sqrt{17})/18$, so no integer family realises both simultaneously; (ii) the leading two-term Birkhoff balance produces no rational-coefficient invariant that distinguishes the Trans locus beyond the (Vieta-tautological) ratio identity; (iii) the sub-leading $[n^{-1}]$ Birkhoff balance is regular on the Trans locus, with the candidate resonance coefficient factoring as $(a_2 - b_1\mu)(4a_2 - b_1^2)$ and vanishing only on the algebraic-discriminant-zero locus, never on Trans.

The combined picture is that the Trans property of degree-$(2,1)$ PCFs is empirically robust, algebraically fingerprinted by $\sqrt{17}$, and not accessible to leading-order or single-step sub-leading Birkhoff methods. We pose the conjecture as a problem in the Galois theory of linear difference equations and in isomonodromic deformation theory.

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1. Introduction

A *polynomial continued fraction (PCF) of degree* (d, e) *with integer coefficients* is an expression

$$K(a,b) \;=\; b_0 \;+\; \cfrac{a(1)}{b(1) \;+\; \cfrac{a(2)}{b(2) \;+\; \cdots}}, \tag{1}$$

where $a(n) = a_d n^d + \cdots + a_0$ and $b(n) = b_e n^e + \cdots + b_0$ are integer polynomials with $a_d \neq 0$ and $b_e \neq 0$. The case $(d, e) = (2, 1)$ is the smallest non-trivial case in which classical limits such as π, Brouncker's $4/\pi$, and Apéry-like series all appear. Writing $a(n) = a_2 n^2 + a_1 n + a_0$ and $b(n) = b_1 n + b_0$, we set

$$r \;:=\; \frac{a_2}{b_1^2}$$

and call this ratio the *Worpitzky parameter* of K.

The companion note [1] announces the *Transcendental Ratio Identity*: every degree-$(2, 1)$ integer PCF whose limit lies in the transcendental stratum (§2) and whose Worpitzky parameter is in the negative interior $(-1/4, 0)$ satisfies $r = -2/9$. The conjecture is supported by zero counterexamples in approximately 585,000 integer families surveyed across the $\mathrm{F}(2, 4)$ and $\mathrm{F}(2, 5)$ complete censuses and several targeted resonance sweeps at the next admissible rational-indicial ratios; a single Log-stratum anomaly at $r = -1/36$ (two families, $b_1 = \pm 6$) is documented and falls outside the conjecture's hypothesis.

The present paper is the long companion of [1]. It extends the announcement in two structurally distinct directions:

- **An algebraic signature (§4).** Under the alternative minus-sign recurrence form $y_{n+1} = b(n)y_n - a(n)y_{n-1}$, the characteristic equation is $\mu^2 - b_1\mu + a_2 = 0$, with roots $\mu_\pm = b_1(3 \pm \sqrt{17})/6$ on the Trans locus $r = -2/9$. We prove (Theorem 4.2) that, for $a_2 < 0$ and $b_1 \neq 0$,

  $$\frac{a_2}{b_1^2} \;=\; -\frac{2}{9} \quad\Longleftrightarrow\quad \left|\frac{\mu_+}{\mu_-}\right| \;=\; \frac{13 + 3\sqrt{17}}{4}\,,$$

  and verify the identity by PSLQ at dps $= 150$ with the phantom-hit rule applied (Theorem 4.4). This is the first explicit algebraic identification of a quadratic irrationality (the discriminant 17) as an invariant of the Trans property of integer degree-$(2, 1)$ PCFs.
- **Three negative results bounding the analytic landscape (§5).** We carry out, in turn, the indicial-exponent forcing (§5.1), the leading-order Birkhoff invariant search (§5.2), and the sub-leading $[n^{-1}]$ balance (§5.3). All three return *negative* answers in a precise sense: no constraint accessible to these standard methods singles out the Trans locus among the admissible rational-indicial loci. The mechanism that produces the empirical universality of $-2/9$ must therefore be *global* rather than local-asymptotic.

The two axes are not independent: $\sqrt{17}$ appears in three guises across the paper — as the discriminant of the characteristic equation on the Trans locus (§4), as the irrational ratio $b_0/b_1 = (27 \pm \sqrt{17})/18$ that obstructs the integer realisation of the indicial pair $\{1/3, 2/3\}$ in minus-sign form (§5.1), and as a coefficient $\mp\sqrt{17}\, b_1/3$ in the regular sub-leading balance on the Trans locus (§5.3). The recurrence of $\sqrt{17}$ in distinct asymptotic settings is, we argue, the algebraic shadow of the global mechanism that the paper otherwise fails to capture.

Organisation. §2 reviews degree-$(2,1)$ PCFs, fixes notation, and contrasts the two recurrence sign conventions in use. §3 restates the Transcendental Ratio Identity and summarises the empirical evidence from [1] (585 000 families, the $\mathrm{F}(2,5)$ census, the resonance sweeps at $|b_1| \le 12$, and the Log anomaly at $r = -1/36$). §4 proves the $|\mu_+/\mu_-|$-modulus identity and gives the PSLQ certificate. §5 contains the three negative results. §6 discusses the global / local dichotomy, the role of $\sqrt{17}$, and four open problems. Appendix A fixes the PSLQ methodology and the phantom-hit rule. Appendix B catalogues the SHA-256 hashes that anchor the AEAL claim manifest.

2. Background and notation

2.1. **Degree-$(2,1)$ PCFs and their strata.** Throughout, $K(a,b)$ denotes a PCF of the form (1) with $a(n) = a_2n^2 + a_1n + a_0$, $b(n) = b_1n + b_0$, and $(a_0, a_1, a_2, b_0, b_1) \in \mathbb{Z}^5$ with $a_2b_1 \neq 0$. We write $\mathcal{F}(2,D) := \{(a_0, a_1, a_2, b_0, b_1) \in \mathbb{Z}^5 : |a_i|, |b_j| \le D,\ a_2b_1 \neq 0\}$ for the parameter cube of side D.

The arithmetic strata used in this paper follow the conventions of [1, 2]. A convergent PCF K with limit L is classified as Rat, Alg, Log, Trans, Des according as $L \in \mathbb{Q}$; L is algebraic of degree ≥ 2; L is a $\mathbb{Q}$-linear combination of logarithms of algebraic numbers; L is transcendental and not in the previous classes (typically a Möbius transform of π or an Apéry-like series); or the partial fractions fail to converge to a finite limit. PSLQ-based stratum identification is performed at dps $= 100$ with 1500 partial quotients and a between-truncant tolerance of 10^{-8}, calibrated against a dps $= 150$ deep-validation pass.

2.2. **Two recurrence conventions.** The companion note [1] uses the *plus-form* recurrence

$$y_{n+1} \;=\; b(n)\, y_n \;+\; a(n)\, y_{n-1} \qquad \text{(plus-form)} \tag{2}$$

which is standard for the convergent sequences P_n, Q_n of a CF written as in (1). The asymptotic analysis of §5 below uses instead the *minus-form* recurrence

$$y_{n+1} \;=\; b(n)\, y_n \;-\; a(n)\, y_{n-1} \qquad \text{(minus-form)} \tag{3}$$

which arises when the sign of a is absorbed into the recurrence relation rather than into the coefficient. The two are conjugate by $y_n \mapsto (-1)^n y_n$ *up to* the

TABLE 1. Sign-convention dictionary on the Trans locus $r = -2/9$.

Form	Recurrence	Eigenstructure on $r = -2/9$
plus	$y_{n+1} = b(n)y_n + a(n)y_{n-1}$	indicial pair $\{1/3, 2/3\} \subset \mathbb{Q}$
minus	$y_{n+1} = b(n)y_n - a(n)y_{n-1}$	char. roots $\mu_\pm = b_1(3 \pm \sqrt{17})/6 \notin \mathbb{Q}$

sign of $a(n)$; the locus $r = a_2/b_1^2$ is invariant, but the indicial exponents and characteristic roots transform. Specifically, on the locus $r = -2/9$:

- The plus-form indicial equation $\lambda^2 - \lambda - r = 0$ has rational roots $\{1/3, 2/3\}$ (Lemma 3.2 of [1], restated here as Theorem 3.1).
- The minus-form characteristic equation $\mu^2 - b_1\mu + a_2 = 0$ has irrational roots $\mu_\pm = b_1(3 \pm \sqrt{17})/6$ (Lemma 4.1).

We use the plus-form for the definitional content of §3 and the minus-form for the asymptotic analysis of §§4–5. A summary correspondence appears in Table 1.

2.3. **PSLQ and the phantom-hit rule.** PSLQ integer-relation detection is performed using `mpmath.pslq` (Python 3, mpmath 1.3.0). Working precision is dps = 100 for census passes, dps = 150 for the algebraic signature certificate of §4, and dps = 200–300 for the sub-leading verification of §5.3. Tolerance is fixed at 10^{-50} and the integer-coefficient bound is set per task as documented in Appendix A.

Phantom-hit rule. Following the convention of the T2A relay pipeline [1], every PSLQ output relation is screened for a *phantom hit*: if the integer coefficient on the target value (the quantity whose closed form is being tested) is zero, the relation is discarded as a phantom of the search basis. In the T2A pipeline, 38 candidate relations were rejected by this rule. The certificates recorded in this paper are explicitly labelled with the target index and the relevant non-zero coefficient.

3. THE CONJECTURE AND EMPIRICAL EVIDENCE

We restate the central conjecture and summarise the empirical evidence from [1]; the reader familiar with the announcement note may skip to §4.

3.1. The plus-form indicial equation and the conjecture.

Theorem 3.1 (Plus-form indicial pair on $r = -2/9$)**.** *Let $K(a, b)$ be a degree-$(2, 1)$ PCF satisfying $a_2/b_1^2 = -2/9$. Then the plus-form recurrence (2) has indicial equation $\lambda^2 - \lambda - r = 0$ with rational root pair $\{\lambda_-, \lambda_+\} = \{1/3, 2/3\}$.*

Proof. With $r = -2/9$, the indicial equation is $\lambda^2 - \lambda + 2/9 = 0$, whose discriminant is $1 - 8/9 = 1/9$, giving roots $\lambda = (1 \pm 1/3)/2 \in \{1/3, 2/3\}$. See [1, Lemma 2.2] for the derivation of the indicial equation itself. □

Conjecture 3.2 (Transcendental Ratio Identity, [1])**.** *Let $K(a, b)$ be a degree-$(2, 1)$ PCF with integer coefficients, $a_2, b_1 \neq 0$, converging to a limit L in*

TABLE 2. Stratum counts in $\mathcal{F}(2,4)$ and $\mathcal{F}(2,5)$ (reproduced from [1]).

Search	Trans	Log	Worpitzky parameter r
$\mathcal{F}(2,4)$ ($D=4$, complete)	24	0	$-2/9$
$\mathcal{F}(2,5)$ ($D=5$, neg. interior)	56	12	$-2/9$
$\mathcal{F}(2,5)$ ($D=5$, Brouncker boundary)	14	0	$+1/4$

TABLE 3. Targeted resonance searches at admissible rational-indicial ratios in the negative Worpitzky interior. All runs at dps $= 100$. Reproduced from [1].

Scope	Convergent families	Trans	Log
$b_1 \in \{\pm 4, \pm 5\}$, $r = -3/16$	2,384	0	0
$b_1 \in \{\pm 4, \pm 5\}$, $r = -4/25$	2,396	0	0
$b_1 \in \{\pm 4, \pm 5\}$, $r = -6/25$	2,394	0	0
$b_1 \in \{\pm 6, \pm 7\}$ (full sweep)	175,686	0	2
$b_1 \in \{\pm 8, \ldots, \pm 12\}$ (motivated targets)	259,280	0	0
Combined	442,140	0	2

TABLE 4. Log-stratum anomaly at $r = -1/36$ (both families have limit $L = \pm 2/\log 2$, dps $= 150$ residual 5.65×10^{-150}).

a_2	a_1	a_0	b_1	b_0
-1	0	0	6	3
-1	0	0	-6	-3

the Trans stratum and with Worpitzky parameter $r = a_2/b_1^2$ in the negative interior $(-1/4, 0)$. Then

$$r \;=\; -\tfrac{2}{9}\,.$$

Equivalently, the plus-form indicial exponents at infinity are $\{1/3, 2/3\}$.

3.2. **Empirical evidence.** Table 2 summarises the four computational experiments of [1]; Table 3 summarises the resonance sweeps; Table 4 catalogues the Log anomaly. No interior-stratum Trans counterexample to Conjecture 3.2 appeared in any of the $\sim$585,000 integer families surveyed.

Remark 3.3 (Phantom-hit discipline). The T2A pipeline used to generate Tables 2–3 applies the phantom-hit rule of §2.3 at every PSLQ stage: 38 candidate relations were rejected during the classification. The two known phantom traps for the bases used here are $\zeta(2) = \pi^2/6$ and the golden ratio $\varphi = (1+\sqrt{5})/2$, arising whenever π or $\sqrt{5}$ enters the basis; neither affected any Trans or Log classification reported above.

3.3. **The Brouncker boundary class.** The 14 families at $r = +1/4$ in $\mathcal{F}(2,5)$ lie on the positive Worpitzky boundary $\Delta = 2$ (where $\Delta = 1 + 4r$ is the plus-form indicial discriminant) and converge logarithmically to Möbius transforms of $4/\pi$. They are excluded from Conjecture 3.2 by the negative-interior hypothesis $r \in (-1/4, 0)$, and do not contradict the Trans Identity. See [1, Theorem 4.1].

4. An algebraic signature: the $|\mu_+/\mu_-|$ identity

We turn to the new content. Throughout this section we work in the *minus-form* recurrence (3).

4.1. The minus-form characteristic equation.

Lemma 4.1 (Minus-form characteristic roots on $r = -2/9$)**.** *Let $K(a,b)$ be a degree-$(2,1)$ PCF, regarded as a minus-form recurrence $y_{n+1} = b(n)y_n - a(n)y_{n-1}$. The Birkhoff ansatz at infinity $y_n \sim \Gamma(n+1)\,\mu^n n^\alpha$ (§5.1 below) gives the characteristic equation*

$$\mu^2 \;-\; b_1\,\mu \;+\; a_2 \;=\; 0 \tag{4}$$

with roots $\mu_\pm = \frac{1}{2}(b_1 \pm \sqrt{b_1^2 - 4a_2})$. In particular, on the locus $a_2/b_1^2 = -2/9$ with $a_2 < 0$ and $b_1 \neq 0$,

$$\mu_\pm \;=\; \frac{b_1\,(3 \pm \sqrt{17})}{6}\,,$$

both irrational quadratic over $\mathbb{Q}$.

Proof. Substituting $y_n = \Gamma(n+1)\mu^n n^\alpha$ into (3), expanding $\Gamma(n+2)/\Gamma(n+1) = n+1$ and $(1 \pm 1/n)^\alpha = 1 \pm \alpha/n + O(1/n^2)$, multiplying through by n^{-2}, and reading the leading $[n^1]$ coefficient (after cancellation against the Γ-recursion) gives the relation $b_1\mu = \mu^2 + a_2$, i.e. (4). See [3, §2] for the full sympy verification. Substituting $a_2 = -2b_1^2/9$ into the discriminant gives $b_1^2 - 4a_2 = b_1^2(1 + 8/9) = 17b_1^2/9$ and $\sqrt{b_1^2 - 4a_2} = |b_1|\sqrt{17}/3$. □

Theorem 4.2 (Algebraic signature of the Trans locus)**.** *Let $K(a,b)$ be a degree-$(2,1)$ PCF with $a_2 < 0$ and $b_1 \neq 0$, and let $\mu_\pm$ be the minus-form characteristic roots of (4). Set $\rho := |\mu_+/\mu_-|$. Then*

$$\frac{a_2}{b_1^2} \;=\; -\frac{2}{9} \quad\Longleftrightarrow\quad \rho \;=\; \frac{13 + 3\sqrt{17}}{4}\,. \tag{5}$$

Proof. Set $q := \mu_+/\mu_-$ (so $\rho = |q|$). Vieta's formulas applied to (4) give $\mu_+ + \mu_- = b_1$ and $\mu_+\mu_- = a_2$, whence

$$\frac{a_2}{b_1^2} \;=\; \frac{\mu_+\mu_-}{(\mu_+ + \mu_-)^2} \;=\; \frac{q}{(1+q)^2}\,.$$

($\Rightarrow$) Substituting $a_2/b_1^2 = -2/9$ gives the quadratic $2q^2 + 13q + 2 = 0$, with roots $q = (-13 \pm 3\sqrt{17})/4$. Both roots are negative; their moduli are $|q| = (13 + 3\sqrt{17})/4$ (for the root with $|q| > 1$) and $|q| = (13 - 3\sqrt{17})/4 =$

$4/(13+3\sqrt{17})$ (the reciprocal). Either choice of labelling μ_+ versus μ_- yields $\rho = (13+3\sqrt{17})/4$ after taking the dominant root in absolute value.
($\Leftarrow$) Conversely, if $\rho = (13+3\sqrt{17})/4$, then $q \in \{-(13+3\sqrt{17})/4,\ +(13+3\sqrt{17})/4\}$ (two sign choices for the quotient). Substituting into $a_2/b_1^2 = q/(1+q)^2$:

- $q = -(13+3\sqrt{17})/4$ gives $a_2/b_1^2 = -2/9$;
- $q = +(13+3\sqrt{17})/4$ gives $a_2/b_1^2 = +2/17$ (positive Worpitzky region).

The hypothesis $a_2 < 0$ excludes the second branch. □

Remark 4.3 (Vieta-tautological character and its content). Theorem 4.2 is, formally, a Vieta tautology: the ratio $|\mu_+/\mu_-|$ is a function of a_2/b_1^2 alone. What is *non-tautological* is that the specific irrational $(13+3\sqrt{17})/4$ on the right-hand side displays the discriminant 17 explicitly. As we shall see in §§5.1–5.3, $\sqrt{17}$ recurs in two further asymptotic settings, and is the algebraic shadow of the Trans property under all three local probes that fail to characterise it.

4.2. **PSLQ certificate.**

Theorem 4.4 (PSLQ certificate)**.** *At working precision* dps $= 150$ *and tolerance* 10^{-50}*:*

(1) *PSLQ on the basis* $\{1, \sqrt{17}, \rho\}$ *with* $\rho = |\mu_+/\mu_-|$ *on a representative Trans family returns the integer relation* $[13, 3, -4]$*, i.e.* $13 + 3\sqrt{17} - 4\rho = 0$*. The target coefficient (on* ρ*) is* $-4 \neq 0$*; the phantom-hit rule passes. Reconstruction error vanishes to* > 100 *decimal digits.*
(2) *PSLQ on the wider basis* $\{1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{17}, \pi, \log 2, \rho\}$ *returns* $[-13, 0, 0, 0, -3, 0, 0, 4]$*, the same identity with* ρ*-coefficient* $4 \neq 0$*.*

Computational verification. Script `forcing_condition.py` of session `2026-04-27/SPRINT-W2-TRANS-INDICIAL-SURVEY/sprint_w2/`. See Appendix B for the SHA-256 hash of the run log. □

5. Three negative results

We now record three negative results. In each, a standard local asymptotic method — (i) indicial-exponent forcing, (ii) leading-order Birkhoff invariant search, (iii) sub-leading $[n^{-1}]$ resonance — is shown to fail to single out the Trans locus among admissible rational-indicial loci.

5.1. **Indicial exponent forcing under the minus form.**

The W1 sprint of this paper [3] works in the minus-form (3) and asks: which integer parameter tuples $(a_0, a_1, a_2, b_0, b_1)$ realise the indicial pair $\{1/3, 2/3\}$ at infinity *in minus-form*? Note this is a different question from Theorem 3.1: in plus-form, $\{1/3, 2/3\}$ is a *consequence* of $r = -2/9$; in minus-form, the corresponding pair on the Trans locus is irrational by Lemma 4.1, and asking for the rational pair $\{1/3, 2/3\}$ is a non-trivial constraint.

Proposition 5.1 (Minus-form indicial closed form, [3])**.** *Under the Birkhoff ansatz $y_n \sim \Gamma(n+1)\,\mu^n n^\alpha$ applied to (3), the next-to-leading $[n^0]$ balance, after using $\mu^2 = b_1\mu - a_2$, has the unique solution*

$$\alpha(\mu) \;=\; -\,\frac{(b_1-b_0)\mu \;+\; (a_1-a_2)}{b_1\mu - 2a_2}\,. \tag{6}$$

Each minus-form characteristic root $\mu_\pm$ produces an indicial exponent $\alpha_\pm := \alpha(\mu_\pm)$.

Proof. Direct sympy substitution and verification; see [3, §3], script `indicial_symbolic.py`. □

Lemma 5.2 (Compatibility ideal for the pair $\{1/3, 2/3\}$)**.** *The pair $\{\alpha_-, \alpha_+\} = \{1/3, 2/3\}$, i.e. the sum-and-product conditions $S = \alpha_+ + \alpha_- = 1$ and $P = \alpha_+\alpha_- = 2/9$, is equivalent to the system*

$$\begin{cases} a_1 - 2a_2 = 0, \\ a_2 \;+\; 9b_0^2 - 27b_0b_1 + 20b_1^2 \;=\; 0, \end{cases}$$

defining the compatibility ideal $C \subset \mathbb{Q}[a_0, a_1, a_2, b_0, b_1]$.

Proof. Substitute (6) into the symmetric functions S and P and simplify using Vieta on (μ_+, μ_-); see [3, §4]. □

Theorem 5.3 (Integer-grid obstruction, negative result)**.** *Let $r = -2/9$. The intersection of the Trans locus with C projects onto the constraint*

$$-9r^2 + 27r - 20 \;=\; r, \quad r := b_0/b_1\,,$$

i.e. $81r^2 - 243r + 178 = 0$, with roots $r = (27 \pm \sqrt{17})/18$, both irrational. Consequently no integer-coefficient family realises both the Trans locus $a_2/b_1^2 = -2/9$ and the minus-form indicial pair $\{1/3, 2/3\}$ simultaneously. Direct enumeration on the integer grid $|b_1| \le 30$, $|b_0| \le 30$, $|a_2| \le 200$ ($\sim 7.3 \times 10^5$ triples) confirms zero simultaneous solutions.

Proof. The root computation is elementary; the enumeration is the script `compatibility.py` of [3]. □

Remark 5.4 (Reconciliation with Theorem 3.1)*.* Theorem 5.3 does not contradict Theorem 3.1: the two work in opposite sign conventions and produce consistent but distinct rational pictures. The conjecture 3.2 as stated uses the plus-form, where indicial $\{1/3, 2/3\}$ on the Trans locus is automatic. Theorem 5.3 says, instead, that under the minus-form (which is the natural setting for Birkhoff asymptotics with the $\Gamma(n+1)$ leading factor), the candidate rational indicial pair $\{1/3, 2/3\}$ requires irrational coefficient ratios involving $\sqrt{17}$, and hence is unrealisable over $\mathbb{Z}$. The integer-Trans empirical content of $-2/9$ *is* reflected in the minus-form picture, but only through the algebraic signature $\sqrt{17}$, not as a rational indicial pair.

5.2. **Leading-order Birkhoff invariant search.** The W2 sprint [4] surveys 50 integer Trans families and 20 non-Trans control families, computing for each the minus-form characteristic data $(\mu_\pm, |\mu_+/\mu_-|, \alpha_\pm, S, P, D)$, where $D = (\alpha_+ - \alpha_-)^2$ is the indicial discriminant.

Proposition 5.5 (No new leading-order invariant, negative result)**.** *Among the leading two-term Birkhoff invariants* $(|\mu_+/\mu_-|, S, P, D)$ *on the integer Trans locus* $a_2/b_1^2 = -2/9$*:*

- $|\mu_+/\mu_-|$ *is constant* $= (13 + 3\sqrt{17})/4$ *(Theorem 4.2, Vieta-tautological);*
- S *takes three distinct rational values across the survey (*$\{-3, -1, +1\}$*);*
- P *takes nine distinct values in* $[-2.235, +2.235]$*;*
- D *takes eight distinct values in* $[0.059, 9.94]$*.*

Consequently, no rational-coefficient invariant accessible to the leading two-term Birkhoff balance, beyond the Vieta-tautological $|\mu_+/\mu_-|$ *identity, distinguishes the Trans locus.*

Proof. Computational survey, script `indicial_survey.py` of [4]; the table of values and the variability ranges are reproduced verbatim from the sprint report. □

Remark 5.6 (Class separation by r). The ratio $|\mu_+/\mu_-|$ is, by Vieta, a function of $r = a_2/b_1^2$ alone; it therefore separates the strata Trans, Log, Alg by construction (Trans has the unique value $(13+3\sqrt{17})/4$; Log families take five distinct values; Alg degenerate or complex). This separation is built into the definition of the strata. The content of Proposition 5.5 is that *nothing else* at leading two-term order is constant across Trans.

5.3. **The sub-leading $[n^{-1}]$ balance.** The W3 sprint [5] extends the Birkhoff ansatz to

$$y_n \;\sim\; \Gamma(n+1)\,\mu^n\,n^\alpha\left(1 + \frac{c_1}{n} + \frac{c_2}{n^2} + \cdots\right)$$

and derives the $[n^{-1}]$ balance, linear in c_1 and c_2. The c_2-coefficient of the balance equals $\mu^2 - b_1\mu + a_2$, which vanishes on the characteristic locus by Lemma 4.1; c_2 is therefore not constrained at this order. The c_1-coefficient A, after using $\mu^2 = b_1\mu - a_2$ and substituting the closed form (6) for α, factors as follows.

Proposition 5.7 (Sub-leading c_1 coefficient, [5])**.** *With notation as above, the coefficient of* c_1 *at the* $[n^{-1}]$ *order of the minus-form Birkhoff balance is, after reduction,*

$$A \;=\; \frac{(a_2 - b_1\mu)(4a_2 - b_1^2)}{a_2 b_1 + 2a_2\mu - b_1^2\mu}\,. \tag{7}$$

Proof. Symbolic computation in sympy; see [5, §2] and script `subleading_birkhoff.py`. The numerator $A_{\text{num}} = (a_2 - b_1\mu)(4a_2 - b_1^2)$ is verified directly; the denominator A_{den} is an explicit rational expression in (a_2, b_1, μ) and is non-zero on the Trans locus. □

Corollary 5.8 (No sub-leading resonance on the Trans locus, negative result)**.** *A_{num} vanishes if and only if (i) $a_2 = b_1\mu$, which combined with the characteristic equation (4) forces $\mu = 0$ and is therefore degenerate; or (ii) $4a_2 = b_1^2$, which is the algebraic-stratum (Alg) discriminant-zero locus. Neither contains the Trans locus $a_2/b_1^2 = -2/9$. On the Trans locus,*

$$A\big|_{\mu=\mu_\pm} = \mp\frac{\sqrt{17}\,b_1}{3} \neq 0\,,$$

so the $[n^{-1}]$ balance uniquely determines c_1 as $-B/A$ (with B the c_1-and-c_2-free remainder), and yields no new constraint on the parameters.

Proof. Combine Proposition 5.7 with Lemma 4.1 for the explicit values of A at $\mu_\pm$. □

Proposition 5.9 (Convergence-rate experiment, negative result)**.** *For the minus-form recurrence on a representative panel of 10 Trans, 5 Log, and 5 Alg integer families at* mp.dps $= 200$ *and $N = 600$ iterations, the partial-fraction error $|r_n - L|$ (with $r_n = p_n/q_n$ and L the high-n limit) reaches 10^{-200} at $n \approx 300$ for every Trans and Log family; the classical normalisation $\rho_n := |r_n - L| \cdot q_n^2$ explodes ($\sim 10^{2620}$ at $n = 500$) because $q_n \sim \Gamma(n+1)\mu_+^n$; the natural rate constant*

$$\lim_{n\to\infty} \frac{|r_n - L| \cdot q_n\, q_{n+1}}{|\mu_-/\mu_+|^n}$$

exists but reduces, on the Trans locus, to the Vieta-tautological quantity $|\mu_-/\mu_+| = 4/(13+3\sqrt{17}) = (13-3\sqrt{17})/4$. No new amplitude or coefficient invariant is exposed by the convergence-rate experiment at this precision.

Proof. Script `convergence_rate.py` of [5]. The hyperexponential rate is a direct consequence of the $\Gamma(n+1)$ factor in the dominant Birkhoff solution; the rate *constant* after Γ-removal is the characteristic-root ratio. □

Proposition 5.10 (Direct PSLQ on the Trans limits, negative result)**.** *At* mp.dps $= 300$*, $N = 500$, and integer-coefficient bound 10^4:*

(i) *PSLQ on $\{1, L, L^2, L^3, L^4\}$ for each of 10 representative Trans CF limits L returns no integer relation.*
(ii) *Pairwise PSLQ on $\{1, L_i, L_j, L_iL_j, L_i^2, L_j^2\}$ returns ten relations among the 45 pairs; all ten are tautological consequences of the linear scaling $L(\lambda b_1) = \lambda L(b_1)$ on the sub-family with $(a_0, a_1, b_0) = (0, 0, 0)$, and are not novel algebraic relations between distinct CFs.*
(iii) *PSLQ on $\{1, L, \sqrt{17}, \pi, \log 2\}$ for each of the 10 Trans limits returns no integer relation.*

Proof. Script `direct_algebraic.py` of [5]. The phantom-hit rule is applied at every step; the scale-invariance hits in (ii) are not phantom in the strict sense (the L-coefficient is non-zero) but are recognised tautological by post hoc inspection. □

6. Discussion and open problems

6.1. **Local versus global mechanism.** The combined effect of §§5.1–5.3 is that no *local* asymptotic method — characteristic exponents, leading invariants, sub-leading resonance, convergence rate — distinguishes the Trans locus from other admissible rational-indicial loci to within the precision and search bounds employed. The empirical universality of $r = -2/9$ over $\sim$585,000 integer families is therefore very probably the shadow of a *global* structural property, of a kind not visible in expansions about $n = \infty$ alone.

We see two natural candidate frameworks for capturing this global property.

6.1.1. *Galois theory of linear difference equations.* The minus-form recurrence (3) is a linear difference operator $\Delta = E^2 - b(n)E + a(n)$ on the field $\mathbb{Q}(n)$. The Galois group of the splitting extension of Δ over $\mathbb{Q}(n)$ (in the sense of Picard-Vessiot for difference equations) classifies the formal solutions modulo rational equivalences. Whether the *integer-realisability* condition (i.e. that some pair of solutions of Δ has a transcendental ratio of integer convergents) corresponds to a specific Galois group — and whether that group is detected by $r = -2/9$ — is to our knowledge open. The recurrence of $\sqrt{17}$ across our three local probes suggests that any such Galois-theoretic answer must involve the quadratic field $\mathbb{Q}(\sqrt{17})$.

6.1.2. *Isomonodromic deformation theory.* The plus-form indicial pair $\{1/3, 2/3\}$ matches the Schwarz-list dihedral entry with exponent difference $1/3$ [9]. The companion note [1] speculates on a Schwarz-monodromic mechanism behind the conjecture; in the present paper we strengthen this speculation by noting that the irrationality $\sqrt{17}$ which shadows the Trans property in three asymptotic settings is exactly the discriminant of the minus-form characteristic polynomial on the Trans locus, suggesting a quadratic-field constraint (rather than the rational $\{1/3, 2/3\}$) is the "correct" invariant for isomonodromic considerations. A Painlevé-VI-type interpretation, in which the Trans locus appears as a special isomonodromic stratum of a difference-equation analogue of Painlevé VI with parameters on the Schwarz dihedral, is a natural structural problem (Problem 2 below).

6.2. **The Log anomaly at $r = -1/36$.** The two Log families at $b_1 = \pm 6$, $r = -1/36$ (Table 4) sit at the intersection of (i) the Bauer–Stern doubling parameter of the Log stratum and (ii) the rational-indicial admissibility set $\{1/q, (q-1)/q\}$ for $q = 6$. Whether this is the unique boundary phenomenon, or the tip of a finer Log-stratum classification, is open (Problem 4).

6.3. **Open problems.**

Problem 1 (Galois-of-difference-equations form). Define the Picard-Vessiot Galois group G_K of a degree-$(2, 1)$ integer PCF. Is there a property of G_K that characterises the Trans locus? Equivalently: is the empirical universality of $r = -2/9$ a constraint on the conjugacy class of G_K (e.g. a specific

finite quotient)? We expect any such characterisation to factor through the quadratic field $\mathbb{Q}(\sqrt{17})$.

Problem 2 (Isomonodromic / Painlevé form). Construct (or rule out) a difference-equation analogue of Painlevé VI in which the Trans locus appears as a special isomonodromic stratum at parameters on the Schwarz-dihedral $\{1/3, 2/3\}$. A natural starting point is the Jimbo-Sakai q-analogue.

Problem 3 (Rational-coefficient extension). Does the Transcendental Ratio Identity (Conjecture 3.2) survive when the integer hypothesis on $(a_0, a_1, a_2, b_0, b_1)$ is relaxed to rational?

Problem 4 (Log anomaly classification). Classify all degree-$(2, 1)$ integer PCFs with $a_2/b_1^2 = -1/36$ producing Log limits. Are the two known $b_1 = \pm 6$ families exceptional, or do larger b_1 produce a Log sub-family at this boundary parameter?

7. Conclusion

The Transcendental Ratio Identity for degree-$(2, 1)$ integer PCFs is empirically robust ($\sim$585,000 families surveyed, zero counterexamples), algebraically fingerprinted by the quadratic irrationality $\sqrt{17}$, and not accessible to any of the three standard local asymptotic methods we have examined. The mechanism producing the empirical universality of $r = -2/9$ must be global — plausibly Galois-theoretic or isomonodromic — and lies beyond the scope of the techniques deployed in this paper.

The role of $\sqrt{17}$ is, we believe, the most striking output of the present investigation. It appears as (i) the characteristic-root discriminant of the minus-form recurrence on the Trans locus (Lemma 4.1); (ii) the irrationality of the b_0/b_1 ratios solving the minus-form indicial-pair compatibility ideal (Theorem 5.3); and (iii) the value of the regular sub-leading c_1-coefficient at $\mu = \mu_\pm$ (Corollary 5.8). We conjecture that any future proof of Conjecture 3.2 will route through the quadratic field $\mathbb{Q}(\sqrt{17})$.

We invite specialists in the Galois theory of difference equations and in isomonodromic deformation theory to consider the structural problems posed in §6.

Appendix A. Computational methodology

All computations use Python 3 with `mpmath 1.3.0` and `sympy 1.14.0`. The PSLQ implementation is `mpmath.pslq` with tolerance 10^{-50}. Working precision is set per task:

- dps = 100 for stratum classification across $\mathcal{F}(2, 4)$, $\mathcal{F}(2, 5)$ and the resonance sweeps;
- dps = 150 for the algebraic signature certificate (Theorem 4.4);
- dps = 200 for the convergence-rate experiment (Proposition 5.9);
- dps = 300 for the direct algebraic identification of the Trans CF limits (Proposition 5.10).

Table 5. SHA-256 hashes of the principal scripts for the algebraic-signature and negative-result sections.

Script	SHA-256 (truncated)
`indicial_symbolic.py` (W1)	see W1 `claims.jsonl`
`compatibility.py` (W1)	see W1 `claims.jsonl`
`indicial_survey.py` (W2)	see W2 `claims.jsonl`
`forcing_condition.py` (W2)	see W2 `claims.jsonl`
`subleading_birkhoff.py` (W3)	`08eddada...dcc0b654`
`convergence_rate.py` (W3)	`b7742ebb...d1e4ecb`
`direct_algebraic.py` (W3)	`88b14129...23685188f`

The integer-coefficient bound `maxcoeff` is documented per PSLQ call; for Theorem 4.4 no explicit bound was needed (the relation has tiny coefficients), and for Proposition 5.10 the bound is fixed at 10^4.

The phantom-hit rule of §2.3 is applied at every PSLQ step. All scripts are archived in the bridge repository papanokechi/siarc-relay-bridge, and every quoted numerical claim has an entry in the AEAL claim manifest (`claims.jsonl`) of the relevant session, recording `evidence_type`, `dps`, script filename, and SHA-256 hash of the run log.

Appendix B. SHA-256 hashes for the AEAL manifest

For reproducibility, the SHA-256 hashes of the principal scripts (at the commit underlying this paper) are tabulated in Table 5. Hashes for the W1, W2, and W3 sprints are recorded in the corresponding session `claims.jsonl` files.

Computational and AI-Assistance Disclosure

Reproducibility. All computations underlying this paper are reproducible from publicly archived scripts and datasets. The classification pipelines and the PSLQ certificates are archived at the bridge repository papanokechi/siarc-relay-bridge, in the sessions referenced in [1, 3, 4, 5]. Every numerical claim is logged as a `claims.jsonl` entry with `evidence_type = computation`, the working precision `dps`, the script filename, and a SHA-256 hash of the result file.

Phantom-hit discipline. All PSLQ relations reported in this paper are accompanied by an explicit non-zero target-coefficient check (the phantom-hit rule of §2.3). In the T2A pipeline used for §3, 38 candidate relations were rejected by this rule; the certificates of §4 pass the rule with target coefficients -4 (basis 1) and 4 (basis 2).

AI-assistance disclosure. During the preparation of this work the author used GitHub Copilot (Microsoft) and Anthropic Claude in order to assist with code generation, numerical computations, and manuscript drafting.

After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of this article. GitHub Copilot was used as the execution agent for all enumerations, PSLQ classifications, and symbolic computations; the agent's outputs are deterministic functions of the stated scripts. Anthropic Claude (`claude.ai`) was used as a critique and drafting partner for the manuscript text. All mathematical claims in this paper were verified by the author against the underlying numerical and symbolic data.

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